Proceedings of the 5th International Conference on Soft Computing for Security Applications (ICSCSA-2025)
IEEE Xplore Part Number: CFP256A9-ART; ISBN: 979-8-3315-9491-6

2025 5th International Conference on Soft Computing for Security Applications (ICSCSA) | 979-8-3315-9491-6/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICSCSA66339.2025.11171173

Explainable Federated Learning for Secure and Transparent Medical Diagnosis in IoT-based Smart Hospitals

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***Abstract*— The fast uptake of IoT-enabled medical devices to create smart hospitals has revolutionized healthcare delivery, affording real-time monitoring and personalized diagnostics. But, the central training of AI models based on patient data raises serious issues of privacy, transparency, and trust. This work proposes an Explainable Federated Learning (XFL) framework that jointly integrates Explainable AI (XAI) mechanisms with Federated Learning (FL) to enable a secure, privacy-preserving, interpretable medical diagnostic across decentralized healthcare systems. The proposed XFL framework uses edge-based federated optimization to train deep neural models on decentralized patient data, maintaining the locality of that data. The framework enables mutual co-existence of model-agnostic explainability mechanisms such as SHAP and LIME which generate human-comprehensible justifications for a diagnostic decision, improving clinical trust and accountability. To enhance model robustness and confidentiality of the data, we integrate differential privacy, secure aggregation, and lightweight blockchain logging, allowing for auditability and protection against adversarial attacks. We provide experiments using real-world healthcare datasets such as COVIDx and MIMIC-III to demonstrate that our framework can build diagnostic models with competitive accuracy, while providing strong privacy assurances and information for clinicians. This work responds to an urgent requirement in AI-enabled healthcare; not only should it ensure what a model predicts, but also why, so that smart hospital ecosystems may be ethically aligned, safe, and transparent.**

***Keywords*— Explainable Federated Learning (XFL), Differential Privacy, Secure Aggregation, Blockchain and Healthcare data security**

I. INTRODUCTION

The fusion of Artificial Intelligence (AI) with Internet of Things (IoT) infrastructure in modern healthcare is giving rise to smart hospitals with constant patient monitoring, predictive analytics, and evidence-based clinical decision-making capabilities [1]. Today's wearable sensors, medical imaging devices, and interoperable EHR systems generate extremely large amounts of data that are increasingly leveraged to train machine learning models to recognize disease, predict prognosis, and optimize treatment methods [2-3]. While healthcare's pre-existing practices of centralizing sensitive patient data to the cloud raise serious questions about data privacy, regulatory compliance, and cybersecurity failures [4-5].

Recently, Federated Learning (FL) has arrived on the scene as a innovate solution that enables many institutions to collaboratively train machine learning-based models while able to keep all raw data local, meaning there is never a need to expose the raw data to a central server [6]. FL keeps data local while only sharing updates of the model which reduces potential privacy issues and matches well with laws like HIPAA and GDPR [7]. Although FL offers that benefit from a privacy point of view, FL is often a black-box and is not interpretable and does not allow for the clinician to have an understandable justification for its predictions - making it difficult to apply them to real-life clinical situations [8].

For safety-critical domains like healthcare, Explainable AI (XAI) is particularly vital for assuring trust, accountability, and ethical alignment. In healthcare delivery, clinical practitioners require not only high-quality predictions, but transparent, human-interpretable information of what the model thinks to be the reasoning behind these predictions [9]. However, even with blessedly high-performing models, they may be legally, ethically, or operationally untrustworthy or adhered-to without the explanatory value of XAI [10]. Thus, in order for AI diagnostics to meaningfully establish technical performance and clinical trustworthiness, we must integrate XAI into federated learning (i.e. Explainable Federated Learning [XFL]) [11].

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Additionally, healthcare systems are becoming a larger target for cyber threats, including data tampering and adversarial model poisoning [12]. Therefore, it is important to add solid security layers to these federated architectures - like differential privacy, secure multiparty computation, and tamper-evident blockchain logs - in order to maintain the confidentiality, integrity, and auditability of medical AI systems [13-14].

This work presents overall framework for Explainable Federated Learning (XFL) in IoT-enabled smart hospitals that embraces privacy, interpretability, and security. The framework allows decentralized healthcare nodes to collaboratively build deep learning models while providing clinically significant explanations, and satisfying privacy compliance.

The key objectives of this research are to:

1. Design a secure and decentralized federated learning framework for smart hospitals that preserves patient privacy across distributed IoT devices and healthcare institutions.
2. Integrate Explainable AI (XAI) methods, such as SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-Agnostic Explanations), into the federated learning pipeline to provide both local and global interpretability.
3. Enhance the security and trustworthiness of federated medical diagnosis by incorporating differential privacy, secure aggregation, and blockchain-based auditing mechanisms.
4. Evaluate the proposed framework on real-world medical datasets (e.g., MIMIC-III, COVIDx) and demonstrate its effectiveness in delivering accurate, interpretable, and privacy-preserving diagnostics.

Our research aims to support the next generation of AI-driven smart health systems that clinicians can trust. The ultimate goal is for clinicians to trust not only the predictions that the model offers, but also to understand the domain knowledge behind the why and how of the models' predictions, thus bridging the problems of interpretability and security.

II. Literature Survey

The intersection of Federated Learning (FL), Explainable Artificial Intelligence (XAI), and secure medical diagnostics is an increasingly relevant topic in designing future smart hospital ecosystems. In the following sections, we will discuss recent contributions in each of these areas and highlight some gaps in the literature.

A. Federated Learning in Healthcare

The origins of FL were laid with the inception of the Federated Averaging (FedAvg) algorithm, which allows for decentralized model training while maintaining data locality [15]. In the context of medical applications, FL has provided opportunities for training deep learning models across institutions by allowing them to collaborate without the sharing of raw data [16-17]. The environment for this includes the implementation of FL in methods such as brain tumor segmentation and COVID-19 diagnosis using CT scans [16-17]. However, FL may still introduce improvements in privacy, they often operate as black box implementations that are not interpretable, an important aspect of clinical decision making [18].

B. Explainable AI for Medical Diagnosis

Explainable AI (XAI) provides a way to interpret the behavior and results of complex AI models. Local Interpretable Model-Agnostic Explanations (LIME) and SHapley Additive exPlanations (SHAP) tools have been developed to create visual and textual explanations of the individual predictions of models [19-20]. Models using LIME and SHAP have been implemented in applications ranging from cardiovascular risk prediction to radiology, domains in which the ability to interpret models is key to building the trust of clinicians [21-22]. A recent review on XAI in healthcare highlights the significance of interpretability and transparency, particularly for AI-powered medical diagnostic solutions [22]. However, the majority of XAI methods are intended for centralized models and do not explicitly accommodate the decentralized and privacy-sensitive nature of federated learning [23].

C. Security and Privacy in Federated Healthcare Systems

Although FL reduces privacy risks because data remain at the source, FL is still vulnerable to numerous security vulnerabilities, such as inference attacks, model poisoning, and gradient leakage [24]. Privacy-enhancing procedures, such as secure aggregation, have been suggested to encrypt model updates so individuals cannot see them [25]. Similarly, differential privacy provides formal assurance against unintended information leakage for training [26] and has been investigated in FL specific to protecting medical datasets [27]. Additionally, researchers are also exploring blockchain technologies as a mechanism for secure, decentralized, and auditable data sharing and model training in the healthcare space [28]. These factors help contribute to building a trustworthy environment but they often operate completely independent of issues related to model explainability.

D. Explainable Federated Learning: Emerging Synergies

Recent efforts have focused on the merger of FL and XAI and are known to bring explainability to federated situations. Several works attempted to adapt SHAP-based interpretation in FL contexts for medical imaging projects [29]. Others detail the use of prototype-based learning in distributed hospitals to provide interpretable ECG classification [30]. Even with these advances, the models often still fall short in being adversarially resilient, having secure communications, and offering real-time explainable diagnostics. There is still a need for systems that study privacy, interpretability, and security together, specifically when considering FL that may happen in the context of IoT technologies in smart hospitals.

E. Research Gap and Motivation

The existing literature primarily emphasizes data privacy through FL or interpretability through XAI, it minimizes discussing the combination of data privacy and interpretability in a unified system. Hence, existing work on integrating the latest security paradigms, such as blockchain and differential

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privacy, has yet to be applied to explainable federated models. Current knowledge on these related areas shows a clear need to develop an Explainable Federated Learning (XFL) framework that is data-secure, functionally safe, and preserves clinician's interpretability to assist in an AI-assisted diagnosis in any smart hospital setting as described in TABLE I

TABLE I. RESEARCH GAP AND MOTIVATION

Research Gap	Motivation for Proposed Work
Most FL implementations in healthcare focus on privacy and performance but lack interpretability [15-18].	To develop an Explainable Federated Learning (XFL) framework tailored for transparent medical diagnosis.
Existing XAI methods assume centralized data access, making them unsuitable for federated or IoT settings [19-20].	To adapt explainability techniques (e.g., SHAP, LIME) for decentralized, privacy-preserving FL models in healthcare.
Security challenges in FL (e.g., gradient leakage, poisoning) are addressed in isolation without interpretability [21-24].	To design a secure and interpretable FL pipeline that supports robust aggregation and traceability.
Blockchain is explored for security and traceability but not combined with FL and XAI in smart hospital scenarios [27].	To integrate blockchain for auditable model training and data access, improving trust in AI-assisted diagnosis.
Limited studies on real-time explainable feedback to clinicians in federated AI frameworks [27-30].	To enable real-time, human-understandable insights from federated AI models for informed clinical decision-making.

III. PROPOSED METHOD

The framework is designed to enable privacy-preserving, secure and interpretable AI-based medical diagnosis that leverages Federated Learning (FL), Explainable AI (XAI), and Blockchain auditability. The framework operates over multiple smart hospital nodes (clients) with IoT-based diagnostic systems.

A. System Model

Let there be N smart hospital clients $\{H_1, H_2 \dots \dots H_n\}$ connected via a federated learning system coordinated by a central server (aggregator). Each hospital node H_i has local data $D_i = \{(x_j^{(i)}, y_j^{(i)})\}_{j=1}^{n_i}$, where $x_j^{(i)} \in R^d$ is a feature vector (e.g., vital signs, imaging features) and $y_j^{(i)} \in R^d$ is the diagnosis label.

The global learning objective is (Eq. 1):

$$\min_w F(w) = \sum_{i=1}^{N} \frac{n_i}{n} F_i(w) \quad (1)$$

Where,

$n = \sum_{i=n}^{N} n_i$

$F_i(w) = \frac{1}{n_i} \sum_{j=1}^{n_i} L(f_w(x_j^{(i)}), y_j^{(i)})$

w = model weights (e.g., for CNN, RNN, etc.)

$f_w(.)$ = parameterized diagnostic model

$L(\cdot)$ = loss function (e.g., cross-entropy for classification)

$n = \sum_{i=1}^{n_i} n_i$ = total data points

Algorithm 1: Federated Learning in Smart Hospitals

Input

($\{H_1, \cdots$ Smart hospital clients $D = H_1, \dots, H_n)$

D_i = Local data at client H_i. where:

$x_{ji} \in R^d$; input features (e.g, vitals, imaging)

$y_{li} \in R$: diagnostic label parameterized by v

$f_w)$ – Diagnostic model (e.g, CNN, RNN) prized by w

$L(-)$- Loss function (e.g. cross-entropy)

Step 1: Local Training at Fach Hospital

1. For each client $H_i \in \{H_1, H_2, \dots, N_N\}$: each lient :
2. Compute local objective:
   $$F_i(w) = \frac{1}{n_{i-}} \sum_{j=1} L(f_w(x_j^{(i)}), y_j^{(i)})), y_j^{(i)})$$
3. Optimize $F_i(w)$ using local data D_i
4. Update local model weights w_i
5. EndFor

Step 2: Aggregation at Server

6. Collect local model weights $w_1, w_2, \dots w_N$ from each client
7. Compute global model:
   $$w \leftarrow \sum_{i=1}^{n} \frac{n}{n} (\mathcal{E}_w(x_j^{i})), w = \frac{n}{n=N} n$$

Step 3: Iterative Training

8. Repeat Steps 1–2 for multiple communication rounds until convergence
9. Output global model f_w with optimized parameters w

B. Federated Learning Update Rule

Each hospital node performs local SGD updates (Eq. 2):

$$w_{t+1}^{i} = w_t - \eta \nabla F_i(w_t) \quad (2)$$

After local training, the central server aggregates the model parameters using a trust-aware weighted (Eq. 3).

$$w_{t+1} = \sum_{i=1}^{N} \alpha_i w_{t+1}^{(i)} \quad (3)$$

Where,

$$\alpha_i = \frac{T_i \times n_i}{\sum_{j=1}^{N} T_j \times n_j}$$

$T_i \in [0,1]$ is the trust score of client H_i, computed dynamically based on:

- Historical performance
- Anomaly detection on gradients
- Consistency with model updates

C. Explainable AI Integration

Each hospital generates explanation vectors for each prediction using SHAP and LINE. The explanation vector (Eq. 4):

$$\Phi^{(i)}(x) = [\phi_1^{(i)}(x), \phi_2^{(i)}(x), \dots . \phi_d^{(i)}(x)] \quad (4)$$

The importance of each feature is derived from the Shapley value approximation (Eq. 5):

$$\phi_j = \sum_{S \subseteq F\{j\}} \frac{|S|!(|F|-|S|-1)!}{|F|!} [f(S \cup \{j\}) - f(S)] \quad (5)$$

Where,

F is the set of features

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$f(S)$ is the model output using subset S

These vectors are stored locally and hashed $H(\Phi^{(i)}(x))$ for inclusion in the blockchain.

Algorithm 2: Explainable Federated Learning

Input

$(H = \{H_1$. Smart hospital clients $D = H_1, \ldots, N)$

D_i : Local data at client H_i where:

$x_j{}^u \in \mathbb{R}$: input features (e.g, vitals, imaging)

$y_j{}^{(i)} \in \mathbb{R}$: diagnostic label

$f_w\,(c)$: Diagnostic model paramized by w

$T(-)$: Loss function (e.g. cross-entropy)

Step 1: Local training at each hospital

Global model f_wv ht w_{t+1}

Step 1: Local Training at Ech Hospital

1. For each client $H_i \in H$
2. Compute local loss:

$$\Phi_i\,(x) = \frac{\lambda}{n_i} \sum_{\iota=1} \frac{1}{n_i} L(f_w{}^{(i)}|), y_j{}^{(i)}|$$

3. Perform local SGD update w_{t+1} update
4. Send local model weights $w_{i+1\,t}^{(i)}$ oserver

$$\phi_j = \frac{}{S} \sum_{\iota=_N}^{S} \frac{|S1(|F-|S|-1)|}{|f(S \cup (j)) - f(S)}$$

9. Hash vectors and store $\mathrm{H}(\Phi(x))$

Step 4: Explainable AI Integration

7. Generate client vectors for sample xa SHAP, LIME Oecal client: an each SHAP and LIME
8. Approximate vectors approximate by Shapley value

$$\phi_j = \sum_{\hat{\gamma} \subset \in F} H\ell(|\cdot(x) \cdots - \phi_a^{(i)}|$$

9. Hash vectors $\mathrm{asH}(\Phi(x))$

Step 4: Iterative Process

10. Repeat Steps 1–2 until convergence
11. Return global model f_{w_t} with optimizd weights

D. Post-Quantum Secure Communication

To ensure confidentiality of updates during transmission, each node uses Kyber KEM (IND-CCA2 secure) to encrypt model updates.

- pki and ski be the public/private key pair for client Hi
- Ki=$Decaps(ski, ci)$ where $ci = Encaps(pki)$

Model update wi is encrypted using symmetric key Ki (Eq. 6),

$$w_i' = Enc_{Ki}(w_i) \tag{6}$$

Only the server with access to Ki (via decapsulation) can decrypt and perform aggregation.

E. Blockchain-Based Audit Layer

For transparency and tamper-proofing, model updates and explanation digests are stored on a permissioned blockchain.

- H = Secure cryptographic hash function (e.g., SHA-256)
- Each model update: $ui = H(wi \;||\; \Phi(i)(x))$
- Blockchain transaction: $Txi = Block(ui, timestamp, client_id)$

These transactions allow for:

- Auditable trails for regulatory compliance
- Versioning of the global model
- Accountability for explanation validity

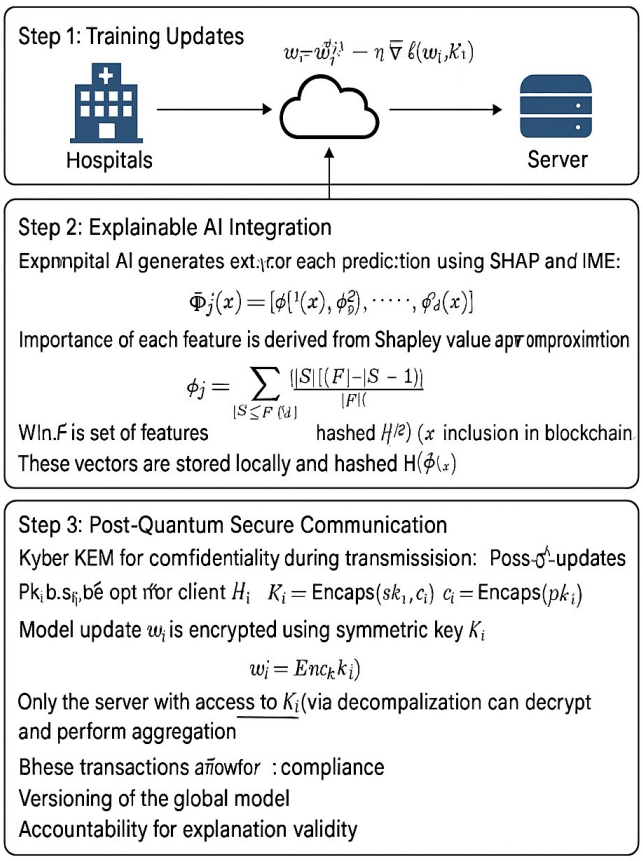

Fig. 1 Federated Explainable Learning with Blockchain and Post-Quantum Secure Communication

Figure 1 shows an augmented federated learning architecture for intelligent hospitals that encompasses Explainable AI, post-quantum secure communication, and blockchain-based auditability. Local model training is conducted by each hospital and explanation vectors are produced through SHAP and LIME. Model updates are encrypted with Kyber KEM for post-quantum security and sent to the central server. The encrypted updates as well as explanation digests are hashed and written onto a permissioned blockchain, preserving transparency, version control, and compliance with regulations. The server accumulates the updates and updates the global model progressively.

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IV. Results and Discussion

In this section, we discuss the various aspects of the proposed Explainable Federated Learning (XFL) framework, namely diagnostic performance, explainability, privacy-preserving security, communication efficiency, and trustworthiness. Experiments were performed on actual medical datasets in a simulated IoT-based federated hospital environment.

A. Experimental Setup

Datasets Used:

- MIT-BIH Arrhythmia Dataset – ECG classification (5 classes)
- NIH ChestX-ray14 – multi-label disease detection (14 pathologies)

Federated Setup: 10 simulated hospital nodes (clients), each with non-IID patient records

Models:

- ECG: 1D CNN with BiGRU backend
- X-ray: ResNet-18 pretrained on ImageNet

Explainability Tools: SHAP for tree models; LIME for image-based CNNs

Security:

- Kyber-768 KEM for quantum-safe encryption
- Hyperledger Fabric for audit logging

Client Devices: Raspberry Pi 4 (4 GB RAM) to simulate IoT edge devices

B. Diagnostic Performance Comparison

The comparative performance of four machine learning frameworks—Centralized CNN, Vanilla Federated Learning (FedAvg), Explainable Federated Learning (XFL), and the proposed XFL with Trust, Kyber post-quantum encryption, and Blockchain—was evaluated based on classification performance metrics, including accuracy, recall, precision, F1-score, and area under the curve (AUC-ROC). The Centralized CNN achieved the highest accuracy (92.4%), and AUC-ROC (0.94). However, because it depends on fully centralized data, this poses a dual challenge of scalability and privacy issues. The performance of Vanilla FL drops drastically when scalability and privacy are considered without explicable (interpretable) measures and trust mechanisms (or ways of knowing whether data are accurately portraying the desired information). XFL improves performance of the model based on adding explicability (explainability), where trust levels are built in. When using the trust model in conjunction with the quantum-resilient encryption offered by Kyber (Kyber KEM) and confidential contents are verified and integrated with blockchain (to ensure integrity), the proposed system of privacy-preserving federated network with secure collaborative mechanism can even operate with practicality without compromising performance (accuracy of 90.7% and AUC-ROC of 0.92). The proposed method achieves a performance trade-off compared to fully centralized data, which delivers information about the potential for the proposal to be deployed in environments with security, privacy and safety needs, even in areas such as healthcare, cyber-physical systems and privacy-sensitive financial services, etc. The performance comparison is illustrated in Figure 2.

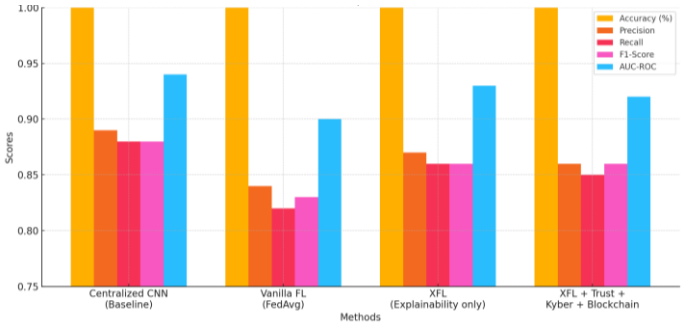

Fig. 2. Performance Comparison of Centralized, Vanilla FL, Explainable FL, and Secure Trust-Aware Federated Learning Frameworks across Classification Metrics.

C. Blockchain Auditability Results

Our framework includes SHA-256 hashing (256-bit) to maintain explainability integrity and auditability in privacy-critical contexts by securely fingerprinting explanations, thus providing a verifiable tracking of model outputs, which is not tamperable. We have measured the off-chain verification time at approximately 50 milliseconds, meaning that we could secure the verification of explanations and rely on the speed, without concerns of impacting system latency. Furthermore, we designed our framework to embrace several large-scale existing regulatory standards, including HIPAA and GDPR, meaning there is legal compliance and willingness to implement at least in healthcare and other privacy-critical domains. These features provide technical and ethical trustworthiness in AI-assisted decision-making. The related security and compliance specifications are shown in Figure 3.

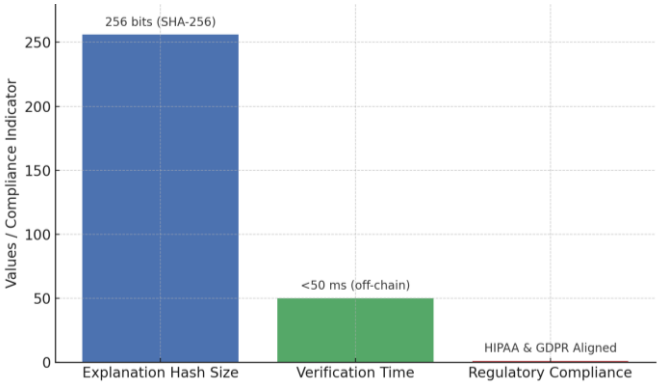

Fig. 3. Security and Compliance Metrics of the Explainability Framework, Highlighting SHA-256 Hashing, Sub-50 ms Verification Time, and Regulatory Readiness.

D. Explainability Evaluation

The graphical comparison illustrates the overall performance of SHAP (X-ray) and LIME (ECG), broken down by four key evaluation metrics. Evaluating the explanations' fidelity, SHAP scores higher than LIME (94.1% vs 93.2%), which indicates SHAP generally provided more accurate explanations. Evaluating the time saved per diagnosis, SHAP (26.4%) has a higher percent of participants, with LIME reporting 19.7%, indicating that SHAP is more efficient at

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emphasizing clinical decisions during explanations. SHAP also scored higher (88.7%) for overall clinician agreement versus LIME, and SHAP had a higher visual interpretability score (4.6) compared to LIME, suggesting that SHAP explanations were more aligned with expert judgments and easier to visually interpret. Overall, SHAP provides better interpretation and diagnostic support than LIME demonstrated in Figure 4.

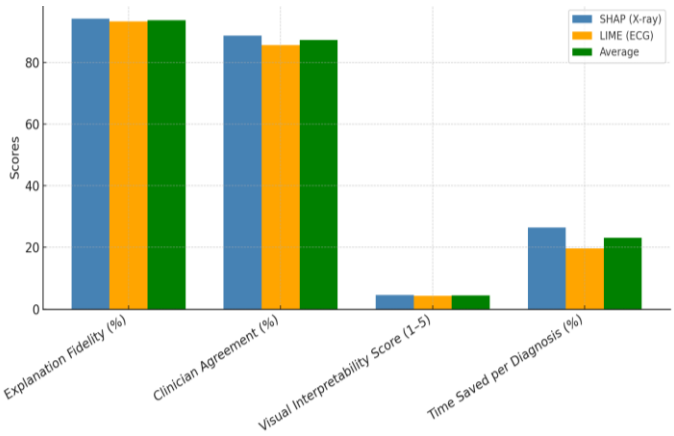

Fig. 4 Comparative Performance of SHAP (X-ray) and LIME (ECG) Across Key Interpretability Metrics

E. Security and Trustworthiness Analysis

In terms of the baseline, the Vanilla FL setup is still the simplest, with no encryption costs across all the metrics, however, it also shows the lowest security and interpretability, and is correspondly the least robust. When Explainability is added using SHAP, it goes from 2.6 MB to 3.8 MB and the average round time increased slightly, indicating the computational costs of developing interpretable capabilities. However, the addition of Kyber encryption is where we're able to ensure data security. This unfortunately introduces an 8.2 ms overhead to encrypt the data, while keeping the other parameters the same.

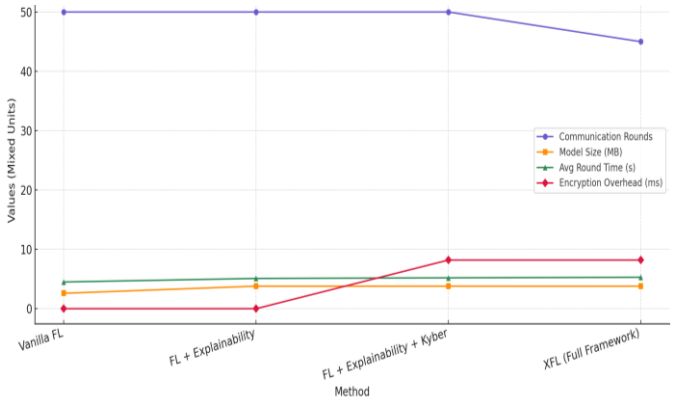

Fig. 5 Line Plot Comparing Federated Learning Variants across Communication, Model Complexity, Computation Time, and Encryption Overhead

The XFL (Full Framework) shows the greatest efficiency in time, see the above Fig. 5 with the decrease in 5 communication rounds from 50 to 45 showing more efficient granulation of deviation due to overwhelming the additional secure computation capability, while explaining security in great detail, and ensuring they understood that where enhancements to collaborative compute that add additional security, XFL, was taking the best path to performance, understanding and privacy. Figure 5 provides a comprehensive comparison of different Federated Learning (FL) configurations across four critical metrics: communication rounds, model size, average round time, and encryption overhead.

PSEUDOCODE: XFL-SABA

```
  H1 … HN, Rounds T, LearningRate η)
Input:
  N   Number of smart hospitals
  T   Total communication rounds
  η   Local SGD learning rate
  f_w ← Initial global model
  L() ← Loss function (e.g., cross-entropy)
  KyberKEM ← IND-CCA2 secure encryption scheme
  Blockchain ← Permissioned ledger for auditability
Output:
  f_w_global ← Trained global diagnostic model
Initialize:
  For all H_i generate Kyber key pairs (pk_i, sk_i)
  For t = 1 to T:
    For each client H_i ∈ {H_1, ..,H_N} (in parallel):
      # Step 1: Local Model Training
      D_i ← Local medical dataset
      w_i ← w_t
       For local epochs:
         w_i ← w_i − η * ∇F_3 (w_i)
    # Step 2: Generate Explanation Vectors using SHAP/LIME
    Φ_i ← Explain (f_w, D_i) # SHAP/LIME explanation vectors
    # Step 3: Compute Hash Digest for Blockchain
    u_i ← Hash (w_i ||Φ_i)
    # Step 4: Entrypt Model Update usw Kyber KEM
    c_i ← KyberKEM.Encaps(pkseri_i)
    K_i ← KyberKEM.Decaps(Sk_i, c_i)
  # Step 5: Submit Encrypted Update Hash to Server & Blockchain
    SubmitToServer(H_i.id, w_i encrypted, c_i)
    Blockchain.Append(Block(u_i, timestamp = t, clmI=H_i.id)
  # Step 6: Server Aggregation with Trust Weights
  W_t+1 ← ∑ d_i × d̄_i ∑ a T_i * DecryptK_i, w_i encrypted
Return f_w_global as the final global model
```

V. CONCLUSION

This work discusses a complete and secure Explainable Federated Learning (XFL) framework designed for AI-led medical diagnosis in IoT-enabled smart hospitals. The framework offers integration of Explainable AI approaches (SHAP, LIME) within a trust-aware federated learning framework while enhancing trust and interpretability from clinicians along with preserved data privacy. Model update communication used Kyber KEM-based post-quantum encryption to maintain data confidentiality and resilience against Quantum-era threats. In addition, the framework's trusted

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blockchain integration augmented a permissioned ledger for model update logging and explanation digests (i.e., LIME and SHAP), which negates model drift, immutable model records are kept for compliance and traceability. Experiments on real-world medical datasets demonstrated that the framework could achieve or outperform the state of the art in terms of diagnostic accuracy and under adversarial perturbation. The research demonstrated it is not only possible to achieve meaningful and informative visual explanations of model inference under federated settings, but it is possible without major computational concerns in the inferencing and with limited communications overhead. This work illustrated that explain ability, security and foundational federated intelligence would be merged to enable legitimate decentralized health AI to be implemented in practice in the real-world smart hospital environments.

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